Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A change of $8.0\ \mathrm{mA}$ in the emitter current bring a change of $7.9\ \mathrm{mA}$ in the collector current. The values of parameters α and β are respectively

(A) $0.99, 90$

(B) $0.96, 79$

(C) $0.97, 99$

(D) $0.99, 79$

Q.2

A working transistor with its three legs marked P, Q and R is tested using a multimeter. No conduction is found between P and Q. By connecting the common (negative) terminal of the multimeter to R and the other (positive) terminal to P or Q, some resistance is seen on the multimeter. Which of the following is true for the transistor?

(A) It is an npn transistor with R as base

(B) It is a pnp transistor with R as base

(C) It is a pnp transistor with R as emitter

(D) It is an npn transistor with R as collector

Q.3

In a common emitter (CE) amplifier having a voltage gain G, the transistor used has transconductance 0.03 mho and current gain 25 . If the above transistor is replaced with another one with transconductance 0.02 mho and current gain 20 , the voltage gain will be

(A) $1.5\mathrm{G}$

(B) $\frac{1}{3}\mathrm{G}$

(C) $\frac{5}{4}\mathrm{G}$

(D) $\frac{2}{3}\mathrm{G}$

Q.4

In a CE transistor amplifier, the audio signal voltage across the collector resistance of $2\mathrm{k}\Omega$ is $2\ \mathrm{V}$. If the base resistance is $1\mathrm{k}\Omega$ and the current amplification of the transistor is 100 , the input signal voltage is :

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(A) $0.1\ \mathrm{V}$

(B) $1.0\ \mathrm{V}$

(C) $1\mathrm{mV}$

(D) $10\mathrm{mV}$

Q.5

In a npn transistor 10^{10} electrons enter the emitter in 10^{-6} s. 4% of the electrons are lost in the base. The current transfer ratio will be

(A) 0.91

(B) 0.93

(C) 0.95

(D) 0.96

Q.6

In common emitter amplifier, the current gain is $62.$ The collector resistance and input resistance are $5\mathrm{k}\Omega$ an 500Ω respectively. If the input voltage is $0.01\ \mathrm{V}$, the output voltage (in volt) is

(A) 6.4

(B) 6.6

(C) 6.9

(D) 6.2

Q.7

The current gain in the common emitter mode of a transistor is $10.$ The input impedance is $20\mathrm{k}\Omega$ and load of resistance is $100\mathrm{k}\Omega$. The power gain is

(A) 400

(B) 300

(C) 600

(D) 500

Q.8

The current gain for a transistor working as common-base amplifier is 0.96. If the emitter current is $7.2\ \mathrm{mA}$, then the base current (in mA) is

(A) 0.23

(B) 0.25

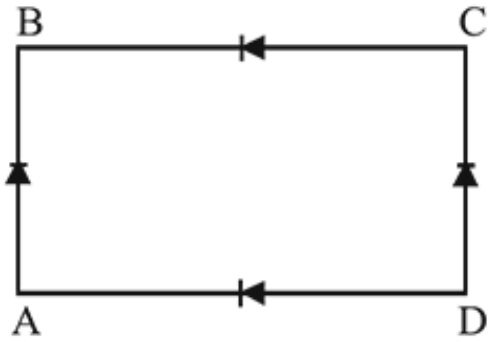
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(C) 0.27

(D) 0.29

Q.9

In fig., the input is across the terminals A and C and the output is across B and D. Then the output is



(A) zero

(B) same as the input

(C) half wave rectified

(D) full wave rectified

Q.10

A npn transistor in a common emitter mode is used as a simple voltage amplifier with a collector current of $4\ \mathrm{mA}$. The terminal of a $8\mathrm{V}$ battery is connected to the collector through a load resistance R_L and to the base through a resistance R_{B}. The collector emitter voltage $V_{CE} = 4V$, base-emitter voltage $V_{\mathrm{BE}} = 0.6$ and the base current amplification factor $\beta_{\mathrm{dc}} = 100$, calculate the value of $\mathrm{R_B}$

(A) $\mathrm{R_B} = 15\mathrm{k}\Omega$

(B) $\mathrm{R_B} = 200\mathrm{k}\Omega$

(C) $\mathrm{R_B} = 1\mathrm{k}\Omega$

(D) $\mathrm{R_B} = 185\mathrm{k}\Omega$

Q.11

The current gain in the common emitter mode of transistor is 10 . The input impedance is $20\mathrm{k}\Omega$ and load of resistance is $100\mathrm{k}\Omega$. The power gain is

(A) 300

(B) 500

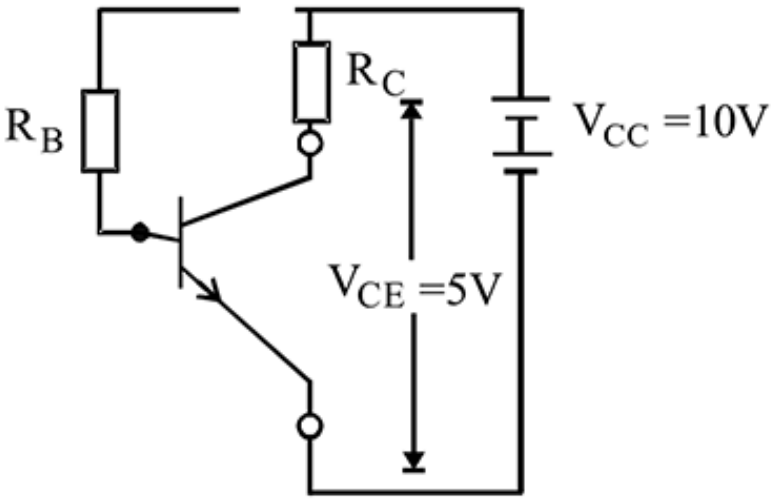
(C) 200

(D) 100

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Q.12

A transistor connected in common emitter configuration has input resistance $\mathrm{R_B} = 2\mathrm{k}\Omega$ and load resistance of $5\mathrm{k}\Omega$. If $\beta = 60$ and an input signal $12\mathrm{mV}$ is applied, calculate the voltage gain, the power gain and the value of output voltage



(A) $A_v = 150, V_{\text{out}} = 1.8\ \text{V}$, and power gain $= 9000$

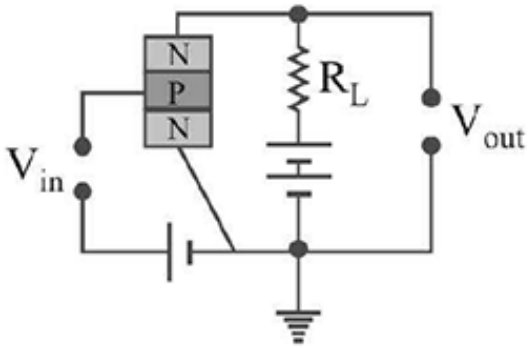
(B) $\mathrm{A_v} = 20,\ \mathrm{V_{out}} = 1\ \text{V}$, and power gain $= 2000$

(C) $A_v = 150, V_{\text{out}} = 1.5V$, and power gain $= 8500$

(D) $A_v = 20, V_{\text{out}} = 1.5V$, and power gain $= 2000$

Q.13

An NPN-transistor circuit is arranged as shown in figure. It is



(A) a common base amplifier circuit

(B) a common emitter amplifier circuit

(C) a common collector amplifier circuit

(D) neither of the above

Q.14

A npn transistor is connected in common emitter configuration in a given amplifier. A load resistance of 800Ω is connected in the collector circuit and the voltage drop across it is $0.8\ \text{V}$. If the current amplification factor

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is 0.96 and the input resistance of the circuit is 192Ω, the voltage gain and the power gain of the amplifier will respectively be :

(A) $4, 3.84$

(B) $3.69, 3.84$

(C) 4,4

(D) $4, 3.69$

Q.15

A transistor can be used as an amplifier when

(A) emitter is forward biased and collector is reverse biased

(B) both are reverse biased

(C) both emitter and collector are forward

(D) None of these

Q.16

In the case of a common emitter transistor amplifier the ratio of the collector current to the emitter current I_c/I_e is $0.96.$ The current gain of the amplifier is

(A) 6

(B) 48

(C) 24

(D) 12

Q.17

In a npn transistor 10^{10} electrons enter the emitter in 10^{-6} s.4% of the electrons are lost in the base. The current transfer ratio will be

(A) 0.98

(B) 0.97

(C) 0.96

(D) 0.94

Q.18

The input resistance of a silicon transistor is 100 W. Base current is changed by $40\mu A$ which results in a change in collector current by 2 mA. This transistor is used as a common emitter amplifier with a load resistance of 4 KΩ. The voltage gain of the amplifier is:

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(A) 2000

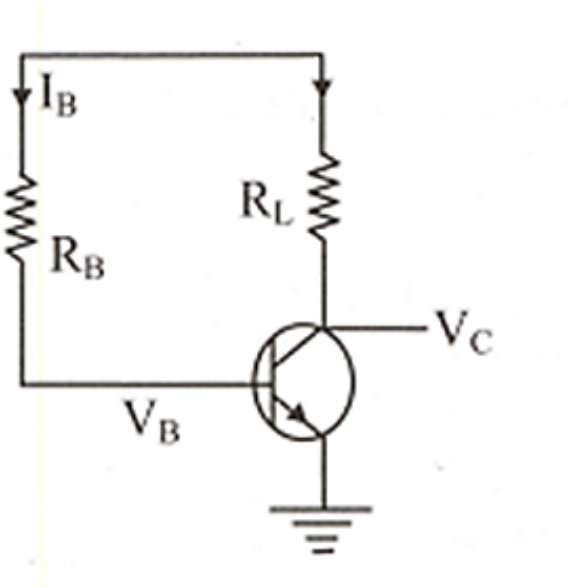
(B) 3000

(C) 4000

(D) 1000

Q.19

In the circuit of CE amplifier, a silicon transistor is used. The value of $V_{CC} = +20$ V, $R_L = 3$ kΩ, collector voltage $= 5$ V, $\beta = 100$. Then the base resistance R_B should be ------ $\times 10^5$ Ω (Take input voltage drop across Si transistor $= 0.7$ V)



(A) 3.76

(B) 3.66

(C) 3.96

(D) 3.86

Q.20

A change of 8.0 mA in the emitter current brings a change of 7.9 mA in the collector current. The values of α and β are

(A) 0.99, 90

(B) 0.96, 79

(C) 0.97, 99

(D) 0.99, 79

Answers & Solutions

Q.1 Answer:

0.99, 79

Solution:

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$\Delta I_E = 8.0\ mA$
$\Delta I_C = 7.9\ mA$
$\alpha = \frac{\Delta I_C}{\Delta I_E} = \frac{7.9}{8.0} = 0.9875 \simeq 0.99$
Also, $\beta = \frac{\alpha}{1-\alpha} = \frac{0.9875}{(1-0.9875)} = 79$

Q.2 Answer:

It is a pnp transistor with R as base

Solution:

It is a $p-n-p$ transistor with R as base.

Q.3 Answer:

$\frac{2}{3}G$

Solution:

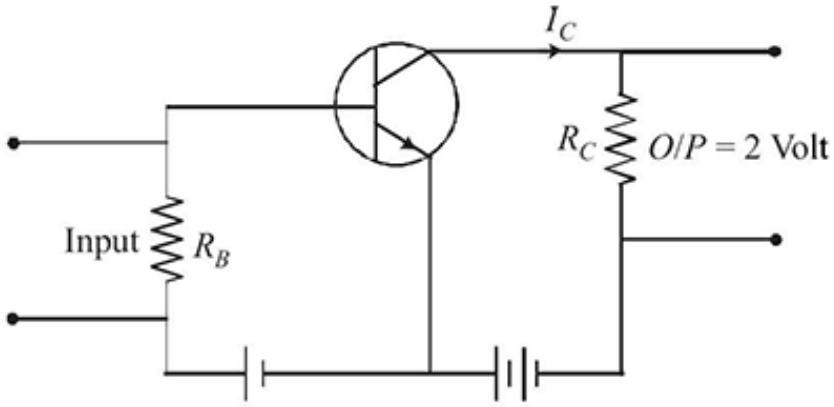
Voltage gain $G = \beta\frac{R_{out}}{R_{in}} \Rightarrow G = 25\frac{R_{out}}{R_{in}}$
Transconductance $g_m = \frac{\beta}{R_{in}}$
$\Rightarrow R_{in} = \frac{\beta}{g_m} = \frac{25}{0.03}$
Putting this value of R_{in} **in eqn. (i)** $G = 25\frac{R_{out}}{25} \times 0.03$
$\therefore \quad G' = 20\frac{R_{out}}{20} \times 0.02$
From eqs. (ii) and (iii) Voltage gain of new transistor $G' = \frac{2}{3}G$

Q.4 Answer:

$10mV$

Solution:

The output voltage, across the load R_C $V_0 = I_C R_C = 2$
The collector current $I_C = \frac{2}{2\times10^3} = 10^{-3}Amp$
Current gain $(\beta) = \frac{I_C}{I_B} = 100$
$I_B = \frac{I_C}{100} = \frac{10^{-3}}{100} = 10^{-5}Amp$
Input voltage (V_i) $V_i = R_B I_B = 1 \times 10^3 \times 10^{-5} = 10^{-2}Volt$
$V_i = 10mV$



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Q.5 Answer:

0.96

Solution:

No. of electrons reaching the collector, $\mathrm{n_C} = \frac{96}{100} \times 10^{10} = 0.96 \times 10^{10}$
Emitter current, $\mathrm{I_E} = \frac{\mathrm{n_E} \times \mathrm{e}}{\mathrm{t}}$
Collector current, $\mathrm{I_C} = \frac{\mathrm{n_C} \times \mathrm{e}}{\mathrm{t}}$
$\therefore$ **Current transfer ratio,** $\alpha = \frac{\mathrm{I_C}}{\mathrm{I_E}} = \frac{\mathrm{n_C}}{\mathrm{n_E}} = \frac{0.96 \times 10^{10}}{10^{10}} = 0.96$

Q.6 Answer:

6.2

Solution:

$\frac{V_0}{V_{\text{in}}} = \frac{R_o}{R_{\text{in}}} \times \beta = \frac{5 \times 10^3 \times 62}{500} = 10 \times 62 = 620$
$V_0 = 620 \times V_{\text{in}} = 620 \times 0.01 = 6.2 \text{ V}$
$\therefore V_0 = 6.2$ **volt.**

Q.7 Answer:

500

Solution:

The power gain in case of CE **amplifier, Power gain** $= \beta^2 \times$ **Resistance gain** $= \beta^2 \times \frac{R_0}{R_i}$
$= (10)^2 \times 5 = 500.$

Q.8 Answer:

0.29

Solution:

Current gain $(\alpha) = 0.96$
$I_e = 7.2 \text{ mA}$
$\frac{I_c}{I_e} = \alpha = 0.96$
$I_c = 0.96 \times 7.2 \text{ mA} = 6.91 \text{ mA}$
$I_e = I_c + I_b$
$\Rightarrow I_b = I_e - I_c = 7.2 - 6.91 = 0.29 \text{ mA}$

Q.9 Answer:

full wave rectified

Solution:

It is the circuit of full wave rectifier.

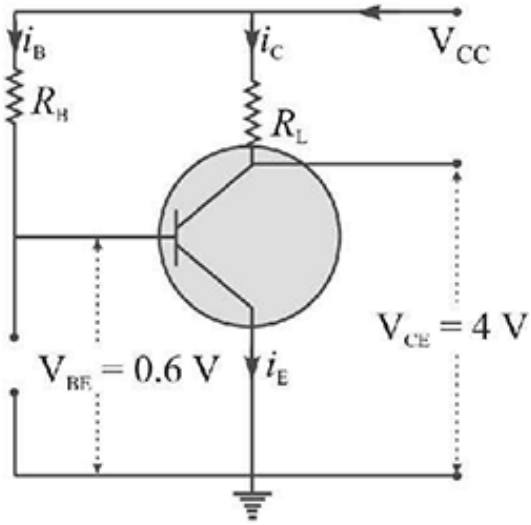
Q.10 Answer:

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$R_B = 185k\Omega$

Solution:

By the definition



$\therefore \quad i_B = \frac{i_C}{\beta_{dc}} = \frac{4mA}{100} = 0.04$ mA

From figure $i_B R_B + V_{BE} = V_{CC}$

$\therefore R_B = \frac{V_{CC} - V_{BE}}{i_B} = \frac{(8-0.6)}{0.04 \times 10^{-3}} = 185k\Omega$

Also $R_L i_C + V_{CE} = V_{CC}$

$\therefore R_L = \frac{V_{CC} - V_{CE}}{i_C} = \frac{8-4}{8 \times 10^{-3}} = 1k\Omega$

Q.11 Answer:

500

Solution:

The power gain in case of CE amplifier, Power gain $= \beta^2 \times$ **Resistance gain** $= \beta^2 \times \frac{R_o}{R_i} = (10)^2 \times 5 = 500.$

Q.12 Answer:

$A_v = 150, V_{\text{out}} = 1.8$ V, and power gain $= 9000$

Solution:

Voltage gain $A_v = \beta \frac{R_C}{R_B} = 60 \times \frac{5 \times 10^3}{2 \times 10^3} = 150$

Power gain $= \beta^2 \frac{R_C}{R_B} = 60 \times 60 \times 2.5 = 9000$

Base current $= I_B = \frac{12 \times 10^{-3}}{2 \times 10^3} = 6 \times 10^{-6}$ A

$I_C = \beta I_B = 60 \times 6 \times 10^{-6} = 3.6 \times 10^{-4}$ A

Output $= I_C R_L = 3.6 \times 10^{-4} \times 5 \times 10^3 \Omega = 1.8$ V

Q.13 Answer:

a common emitter amplifier circuit

Solution:

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In the circuit emitter is common.

Q.14 Answer:

$4, 3.84$

Solution:

$$P_v = A_v\alpha = 4 \times 0.96 = 3.84$$

Q.15 Answer:

emitter is forward biased and collector is reverse biased

Solution:

The emitter produces majority charge carriers, so in order that these reach collector, emitter is forward biased, while collector is reverse biased

Q.16 Answer:

24

Solution:

$$\frac{\mathrm{I_c}}{\mathrm{I_e}} = 0.96 \Rightarrow \mathrm{I_c} = 0.96\mathrm{I_e}$$
$$\text{But } \mathrm{I_e} = \mathrm{I_c} + \mathrm{I_b} = 0.96\mathrm{I_e} + \mathrm{I_b}$$
$$\Rightarrow \mathrm{I_b} = 0.04\mathrm{I_e}$$
$$\therefore \text{ Current gain } \beta = \frac{\mathrm{I_c}}{\mathrm{I_b}} = \frac{0.96\mathrm{I_e}}{0.04\mathrm{I_e}} = 24$$

Q.17 Answer:

0.96

Solution:

No. of electrons reaching the collector, $\mathrm{n_c} = \frac{96}{100} \times 10^{10} = 0.96 \times 10^{10}$

Emitter current, $\mathrm{I_E} = \frac{\mathrm{n_E} \times \mathrm{e}}{\mathrm{t}}$

Collector current, $\mathrm{I_c} = \frac{\mathrm{n_C} \times \mathrm{e}}{\mathrm{t}}$

$\therefore$ Current transfer ratio,

$$\alpha = \frac{\mathrm{I_c}}{\mathrm{I_E}} = \frac{\mathrm{n_C}}{\mathrm{n_E}} = \frac{0.96 \times 10^{10}}{10^{10}} = 0.96$$

Q.18 Answer:

2000

Solution:

Voltage gain $(A_V) = \frac{V_{\text{out}}}{V_{\text{in}}} = \frac{I_{\text{out}}}{I_{\text{in}}} \times \frac{R_{\text{out}}}{R_{\text{in}}}$ $A_V = \frac{2\times10^{-3}}{40\times10^{-6}} \times \frac{4\times10^{3}}{100} = 2 \times 100 = 2000$

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Q.19 Answer:

3.86

Solution:

$V_{CC} = V_C + I_C R_L$

$\therefore 20 = 5 + 3 \times 10^3 I_C$

$\therefore I_C = \frac{15}{3 \times 10^3}$

$= 5 \times 10^{-3} \quad A = 5 \quad mA$,

$\beta = \frac{I_C}{I_B}$

$\therefore I_B = \frac{I_C}{\beta} = \frac{5 \times 10^{-3}}{100}$

$= 5 \times 10^{-5} \quad A$

$V_{CC} = I_B R_B + P.D.$ **across input junction**

As silicon transistor is in forward bias, the $P.D.$ **across input junction is** 0.7 **volt.**

$\therefore 20 = 5 \times 10^{-5} R_B + 0.7$

$\therefore R_B = \frac{19.3}{5 \times 10^{-5}}$

$\therefore R_B = 3.86 \times 10^5 \Omega$

Q.20 Answer:

0.99, 79

Solution:

Given that, change in emitter current, $\Delta I_E = 8 \quad mA$ **and change in collector current,** $\Delta I_C = 7.9 \quad mA$.

We know that,

$\alpha = \frac{\Delta I_C}{\Delta I_E}$

$\Rightarrow \alpha = \frac{7.9}{8}$

$\Rightarrow \alpha \approx 0.99$

Also, we know that

$\beta = \frac{\alpha}{1 - \alpha}$

$\Rightarrow \beta = \frac{\frac{7.9}{8}}{1 - \frac{7.9}{8}} = \frac{7.9}{8 - 7.9}$

$\Rightarrow \beta = \frac{7.9}{0.1} = 79$

Hence, the required answer is $\alpha = 0.99$ **and** $\beta = 79$.

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